

Conic Sections

A cone is a surface generated by rotating a straight line keeping one of its ends in contact with a fixed point and the other in contact with a closed curve. The rotating straight line is called *generator*, the fixed point is called *apex* or *vertex*. The closed curve is called *base*. The base of a circular solid cone is a circle. The line joining the centre of the circle and the apex is called the *axis* of the cone. If the axis of a circular cone is perpendicular to its base, the cone is called *right circular cone*.

If a solid is cut by a plane, a section is obtained. The sections obtained by cutting a right circular cone at different angles, are called *conic sections*. Conic surface is supposed to extend to infinity in both directions from the apex, giving rise to a double cone. Each half of the double cone is called *nappe*. A right circular double cone is shown in Fig. 6.1.

6.1 TYPES OF CONIC SECTIONS

The following are the types of conic sections obtained by cutting a right circular cone by section plane AA, BB, CC, DD, EE, and FF as shown in Fig. 6.1.

1. **Circle:** When a right circular cone is cut by a section plane AAAA, perpendicular to the axis of the cone, the section obtained is called *circle*.
2. **Ellipse:** When a right circular cone is cut by a section plane BBBB, inclined to the axis of the cone and if

the plane cuts all the generators, the section obtained is called *ellipse*.

3. **Parabola:** When a right circular cone is cut by a section plane CCCC, inclined to the axis of the cone and parallel to one of the generators, the section obtained is called *parabola*.
4. **Hyperbola:** When a right circular cone is cut by a section plane DDDD, inclined to the axis by an angle smaller than that the generators make with the axis, the conic section obtained is called *hyperbola*.
5. **Rectangular hyperbola:** When a section plane EEEE, cuts both parts of the double cone on the same side of the axis and parallel to it, the section obtained is called *rectangular hyperbola*.
6. **Isosceles triangle:** When a right circular cone is cut by a section plane through the apex of the cone, the section obtained is called *isosceles triangle*.

For the pictorial view of the above sections see Figs. 6.2 (a) to (f).

6.2 NOMENCLATURE

The concept of important terms used in conic can be better understood by cutting a right circular cone by a tangent plane and a section plane. The section plane may be taken in five

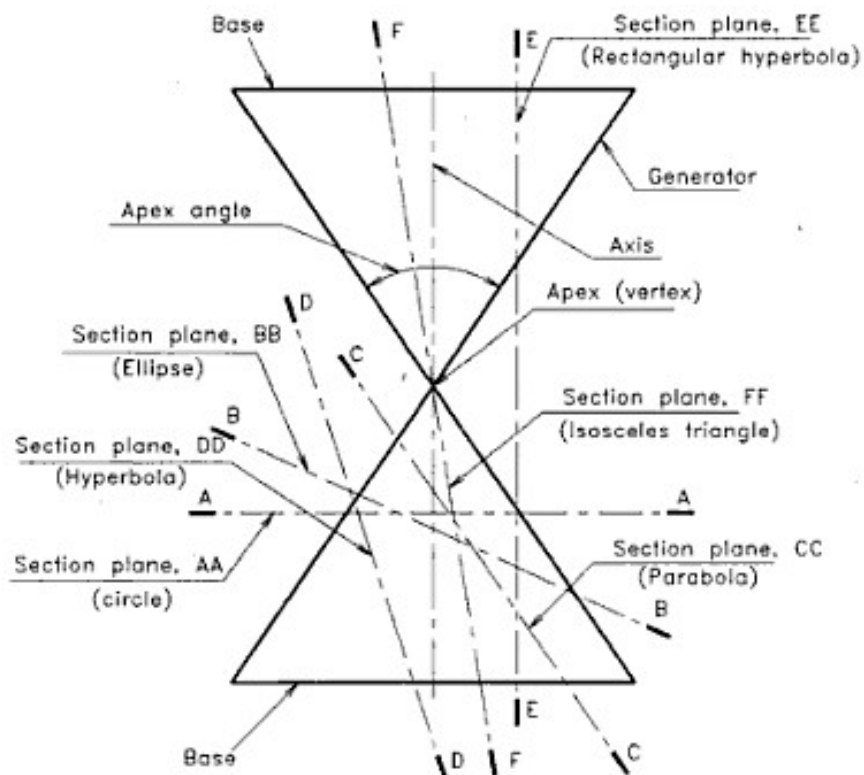


Fig. 6.1 A right circular double cone cut by section planes.

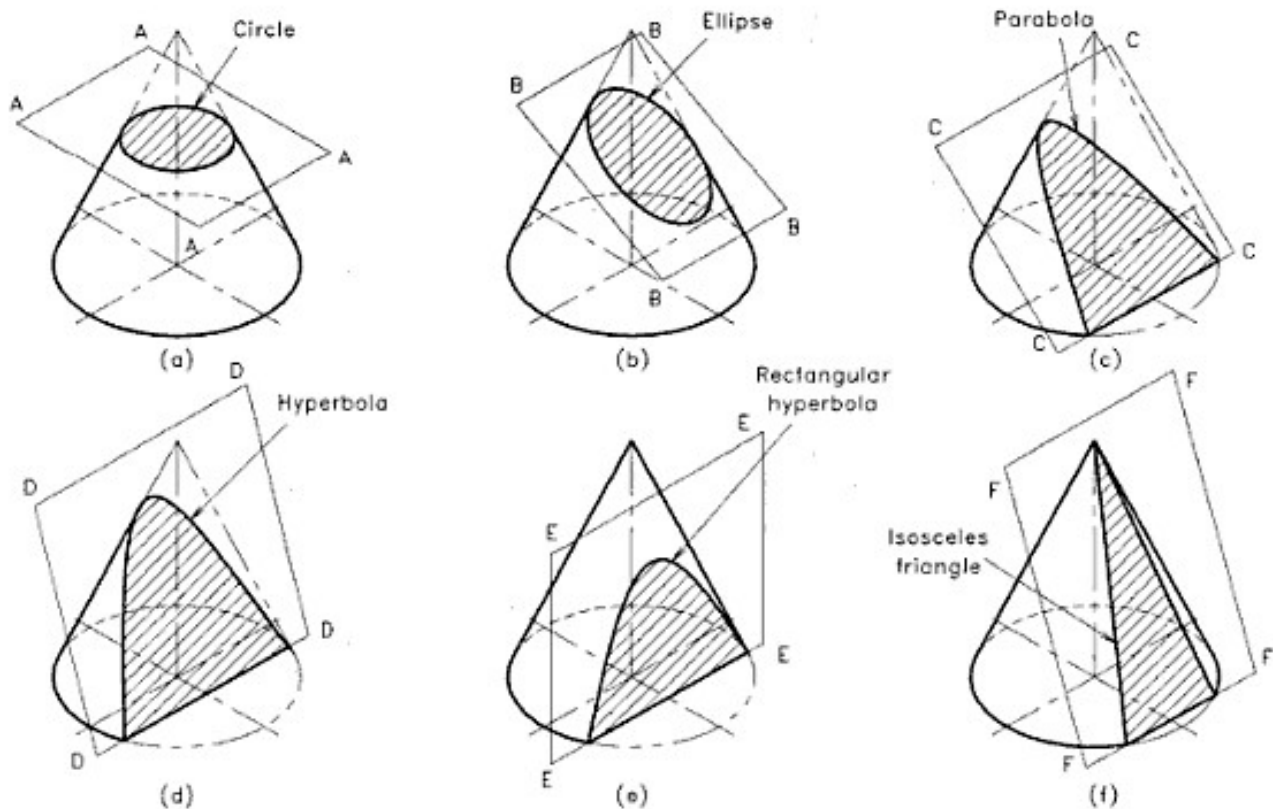


Fig. 6.2 Types of conic sections.

different ways as discussed in the above section. The important terms used in conics are defined below:

1. **Tangent plane:** If a sphere is inscribed in a cone, the sphere touches the cone in a circle. The plane containing the circle is called *tangent plane*. Tangent plane is perpendicular to the axis of the cone.
2. **Focal sphere:** The sphere inscribed in a cone touching the section plane is called *focal sphere*.
3. **Focus of the conic:** If a sphere is inscribed in a cone to touch the section plane, the point of contact between the sphere and the section plane is called *focus of the conic*.
4. **Directrix of the conic:** Tangent plane and section plane meet along a line which is called *directrix of the conic*.
5. **Axis of the conic:** The line passing through the focus and perpendicular to the directrix is called *axis of the conic*. Note that the axis of the cone is different from the axis of the conic.
6. **Vertex of the conic:** The point at which the conic cuts its axis is called *vertex of the conic*.
7. **Chord of a conic:** A straight line joining two points on a conic is called *chord of a conic*.
8. **Focal chord:** A chord which passes through the focus is called *focal chord*.
9. **Ordinate:** The perpendicular from any point on a conic to its axis is called *ordinate*.
10. **Double ordinate:** A length twice that of the ordinate is called *double ordinate*.
11. **Latus rectum:** A double ordinate that passes through the focus is called *latus rectum*.
12. **Abscissa:** The distance along the axis between the vertex of the conic and the point where the ordinate cuts the axis is called *abscissa*.

6.3 CONIC DEFINED AS LOCI

Apart from the definitions given in the above section, a conic can be defined as the path of a point moving in a plane satisfying certain conditions as described below.

A conic may be defined as the locus of a point which moves in a plane such that the ratio of its distance from a fixed point and a fixed straight line in the plane, is a constant. The fixed points are called *focus* and the fixed straight line is called *directrix*. (see Fig. 6.3).

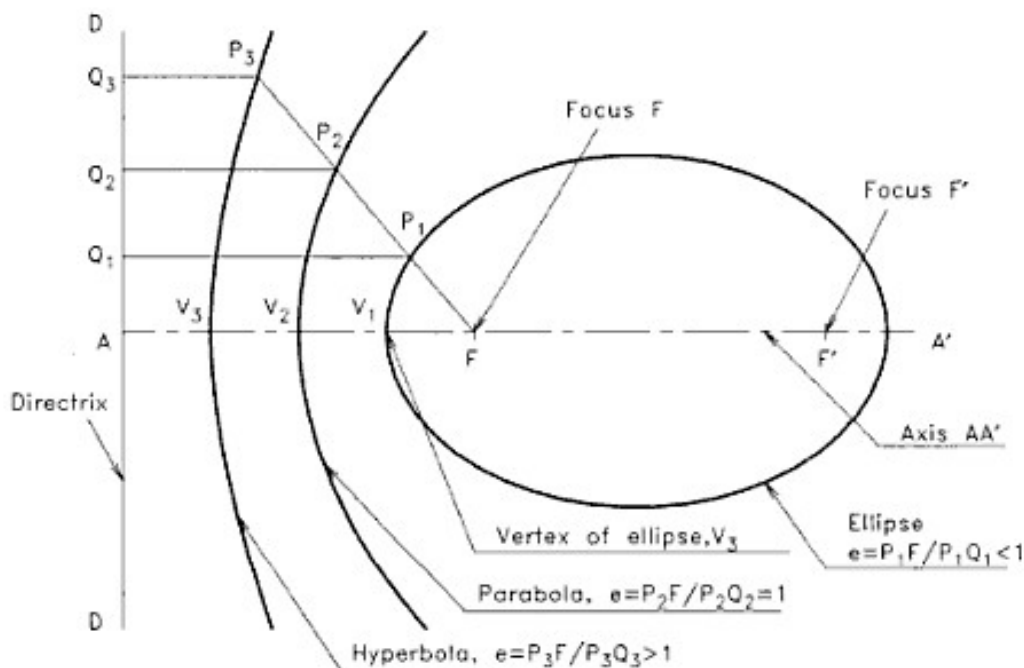


Fig. 6.3 Conic curves defined as loci.

Eccentricity of a conic is the ratio of the distance of the point from the focus to its perpendicular distance from the directrix. It is constant for a conic and is usually denoted by e .

Let P be any point on a curve moving in such a way that the ratio of its distance from a fixed point F and a fixed straight line, DD is constant. PF is the distance of the moving point P from the focus F and PQ is the distance of the moving point P from the directrix.

1. If the ratio $P_1F/P_1Q_1 < 1$, the curve obtained is called *ellipse*.
2. If the ratio $P_2F/P_2Q_2 = 1$, the curve obtained is called *parabola*.
3. If the ratio $P_3F/P_3Q_3 > 1$, the curve obtained is called *hyperbola*.

6.4 ELLIPSE

An ellipse may be defined as a curve traced out by a point moving in a plane such that its distance from a fixed point called *focus* is always less than its distance from a fixed line called *directrix*. An ellipse can also be defined as a curve traced out by a point moving in a plane such that the sum of its distances from two fixed points is a constant. An ellipse has two foci and two directrices (see Fig. 6.4). Since the ellipse cuts the axis of the conic at two points, it is called a *central conic*.

The important terms associated with an ellipse are defined below:

1. **Major axis:** It is the line which passes through the foci and terminated by the ellipse.
2. **Minor axis:** It is the line which passes through the geometrical centre of the ellipse and perpendicular to the major axis and terminated by the ellipse.
3. **Auxiliary circles:** The circles described about the major and minor axes as diameters are called *auxiliary circles*.
4. **Major diameter:** The diameter of the circle described on the major axis of the ellipse is called *major diameter*.
5. **Minor diameter:** The diameter of the circle described as the minor axis of the ellipse is called *minor diameter*.
6. **Tangent to an ellipse:** It is a line which passes through a point on the ellipse and bisects the exterior angle formed by the focal lines at that point. Tangents at the extremities of any diameter are parallel.
7. **Normal to an ellipse:** Normal to an ellipse at any point is the perpendicular to the tangent at that point.
8. **Conjugate diameters:** These are two lines

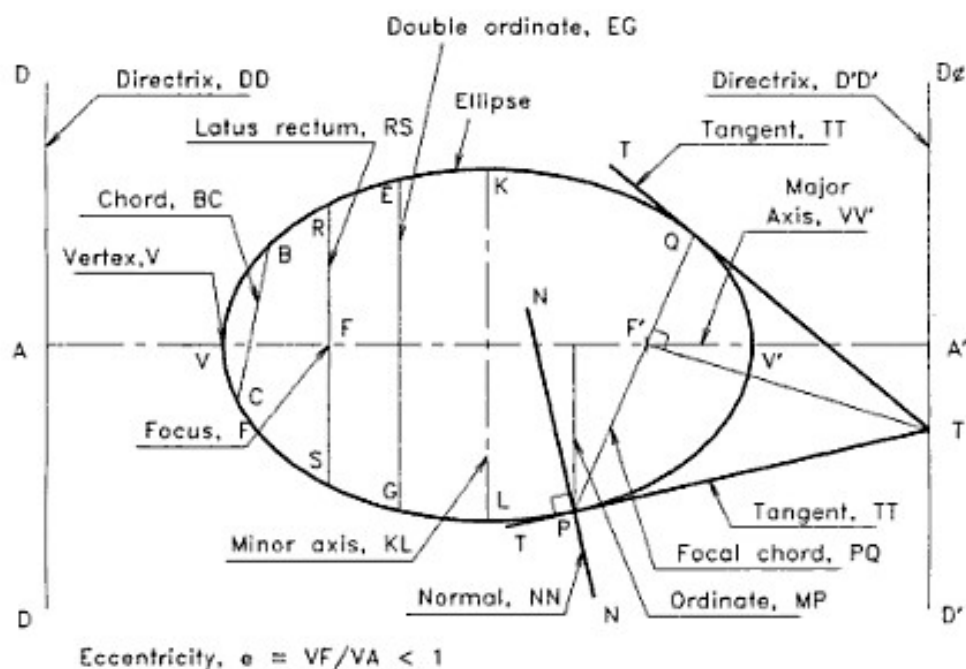


Fig. 6.4 Terminology of ellipse.

intersecting each other at the centre of the ellipse such that each is parallel to the tangents drawn at the extremities of the other.

6.5 METHODS OF CONSTRUCTION OF ELLIPSE

Ellipse is the most commonly used mathematical curve. The elliptical shapes are used for the arches of buildings, in architecture, flanges of machine parts, elliptical gears, drawing pipe end when joining to a plane, etc. There are various methods of constructing an ellipse. A list of the commonly used methods are given below.

1. Eccentricity method
2. Concentric auxiliary circle method
3. Rectangular oblong method
4. Parallelogram method
5. Intersecting arc method
6. Pin and thread method
7. Circle method.
8. Approximate ellipse by four centres.

The data required to construct an ellipse varies with the method of construction. The following examples explain the first six methods of the above list. Procedure of drawing tangent and normal to the curve is also described with them.

Example 6.1

Construct a conic when the distance of its focus from its directrix is equal to 50 mm and its eccentricity is $2/3$. Name the curve, mark its major axis and minor axis. Draw a tangent at any point P on the curve

Refer to Fig. 6.5.

1. As the eccentricity is less than 1, i.e. $e = 2/3 < 1$, the curve is ellipse. Draw one directrix DD and the axis AA perpendicular to DD and mark the focus F such that $FA = 50$ mm.
2. As the eccentricity is $2/3$, divide FA into $2 + 3 = 5$ equal parts. By definition $VF/VA = 2/3$ and hence locate the vertex V.
3. Draw VE perpendicular to the axis such that $VE = VF$. Join AE and extend it as shown in Fig. 6.5. This is the eccentricity scale, which gives the distance directly in the required ratio. In triangle AVE, $VE/VA = VF/VA = 2/3$.
4. Mark any point 1 on the axis and draw a perpendicular through it to intersect AE produced at 1'. With centre F and radius equal to 1-1' draw arcs to intersect the perpendicular through 1 at P_1 , both above and below the axis of the conics.

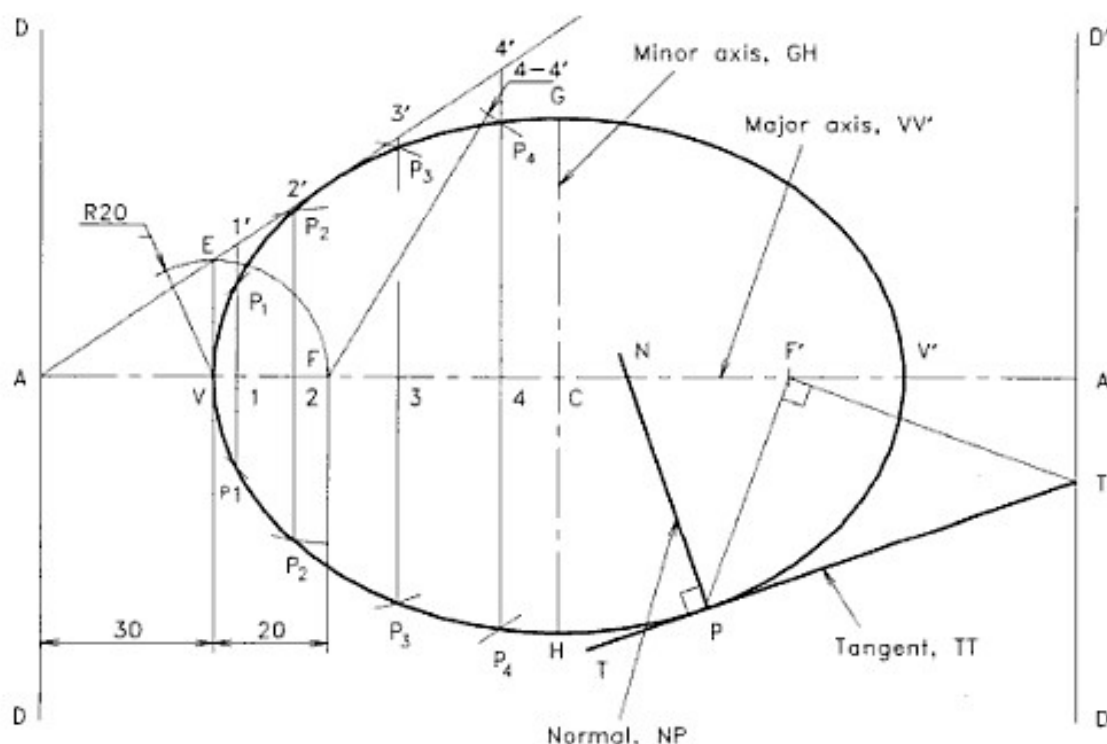


Fig. 6.5 Construction of an ellipse (eccentricity method).

5. Similarly, mark points 2, 3, 4, etc. on the axis and obtain points P_2, P_3, P_4 , etc. as described above. Draw a smooth curve passing through the points V, P_1, P_2, P_3, P_4 etc. which is the required ellipse.
6. Mark the centre C of the ellipse and draw a perpendicular GH to the axis. Also mark the other focus F such that $CF = CF$.
7. Tangent at any point P on the ellipse can be drawn by joining PF' and by drawing $F'T$ perpendicular to PF' . Join TP and extend. Draw NP perpendicular to TP . Now, TPT and NPN are the required tangent and normal at P respectively.

Example 6.2

Draw an ellipse by concentric circles or auxiliary circle method. Given the major and minor axes as 120 mm and 80 mm respectively. Draw a tangent at any point on the ellipse using auxiliary circles.

Refer to Fig. 6.6.

1. With diameters $AB = 120$ mm and $CD = 80$ mm, draw two concentric circles with O as centre. Divide the circle into any number of equal parts (say 12) as shown in Fig. 6.6.
2. Through 2 and $2'$, draw parallels to CD and AB respectively to intersect them at P_2 .
3. Similarly, obtain other points P_3, P_5, P_6 , etc. Point P_1 will be at A itself.
4. Draw a smooth curve passing through all these points to get the required ellipse.

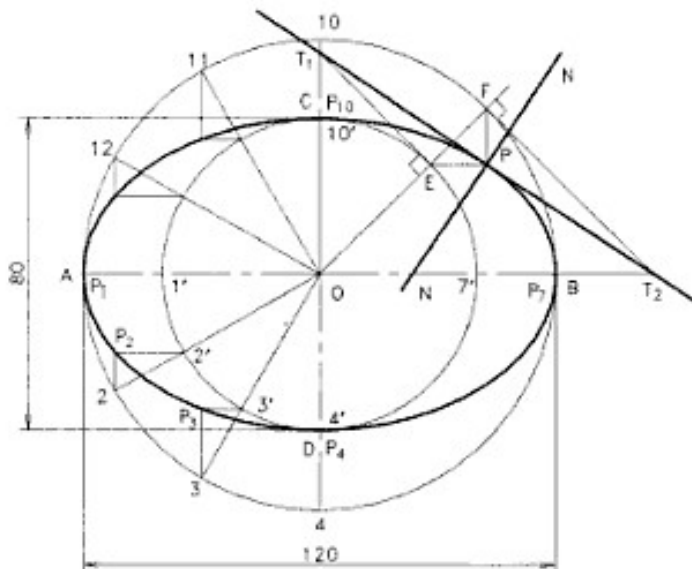


Fig. 6.6 Construction of an ellipse (concentric or auxiliary circle method).

5. Tangent at any point P on the ellipse can be drawn by first locating points E and F . For this, draw horizontal and vertical lines through P to meet the circles at E and F . Draw a radial line OEF . Draw a tangent ET_1 at E to meet the minor diameter at T_1 .
6. Similarly, draw another tangent FT_2 at F to meet the major diameter at T_2 .
7. Join T_1PT_2 and it is the required tangent. Draw NPN perpendicular to T_1PT_2 to get the required normal.

Example 6.3

Draw an ellipse by rectangular method, given the major and minor axes as 150 mm and 90 mm respectively. Draw normal and tangent at any point on the ellipse at a distance of 55 mm from the geometrical centre of the ellipse.

Refer to Fig. 6.7.

1. Draw two axes $AB = 150$ mm and $CD = 90$ mm bisecting each other at O and construct a circumscribing rectangle $EGKL$ as shown in Fig. 6.7.
2. Divide AO and AE into the same number of equal parts (say 4) and name the dividing points. Join C with the points on the side AE .
3. Draw lines from D through points on side AO and extend these lines to intersect the previously drawn lines at P_1 and P_2 . Note that $1C$ intersects with $1'D$ produced at P , and so on.
4. Similarly, obtain points on the other side of the axis. Draw a smooth curve passing through all these points and the curve obtained is the required ellipse.
5. Mark any point P on the ellipse at a distance of 55 mm from the geometrical centre of the ellipse.

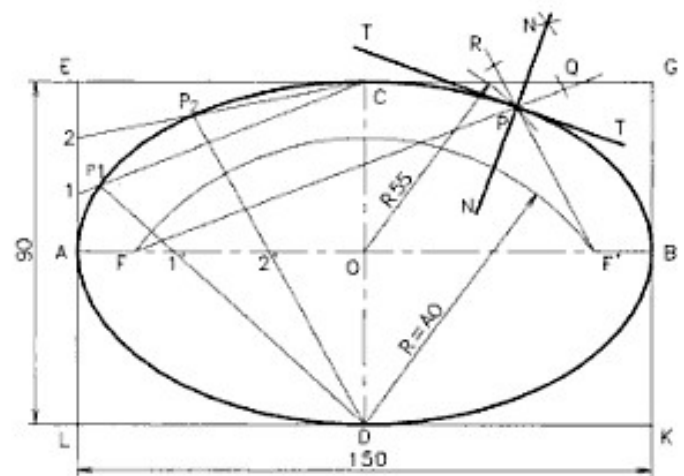


Fig. 6.7 Construction of an ellipse (rectangular or oblong method).

Locate the foci F and F' . Join FP and $F'P$ and produce these lines to Q and R respectively.

6. Draw a line TPT bisecting the angle $F'PQ$. Now, TPT is the required tangent. Draw another line NPN bisecting the angle RPQ . Here, NPN is the required normal.

Example 6.4

Draw an ellipse by the parallelogram method if the conjugate diameters are 150 mm and 108 mm with an included angle of 70° . Also mark the minor axis of the curve.

Refer to Fig. 6.8.

1. Draw the given conjugate diameters $AB = 150$ mm and $CD = 108$ mm with an included angle of 70° . The conjugate diameters bisect at O as shown in Fig. 6.8. Draw the circumscribing parallelogram $EFGH$.
2. Construct the ellipse by rectangular method as shown.
3. Draw a semicircle with O as centre and OD as radius to intersect the ellipse at K . Join CK . The line LM passing through the centre O and parallel to CK is the minor axis.
4. Draw a line NS perpendicular to LM . Here, NS is the major axis.

Example 6.5

The foci of an ellipse are 100 mm apart and the minor axis is 80 mm long. Measure the length of the major axis. Draw the ellipse by using intersecting-arc method.

Also draw a tangent to the ellipse from any point outside the ellipse.

Refer to Fig. 6.9.

1. Draw a horizontal line and mark the foci F and F' on this line such that $FF' = 100$ mm. Draw a perpendicular bisector and mark points C and D such that $CO = DO = 40$ mm.
2. Mark A on the horizontal line such that $AO = CF$. Similarly, obtain the point B . Now AB is the major axis.
3. Mark some points on AO equal to the number of points desired to be plotted in each quadrant of the ellipse.
4. With centres F and F' and radii equal to $1A$ and $1B$ respectively draw arcs to intersect at P_1 on either side of the major axis. Similarly with centres F and F' and radii equal to $1B$ and $1A$ respectively draw arcs to intersect at points P_1 on the right hand side of the minor axis.
5. Repeat the procedure explained above and get the corresponding points P_2, P_3 , etc.
6. Draw a smooth curve passing through all these points using french curves and the curve obtained is the required ellipse.
7. Mark any point P outside the ellipse. With F as centre and AB as radius draw an arc GH . With P as centre and PF' as radius draw another arc to intersect the arc GH at points G and H .

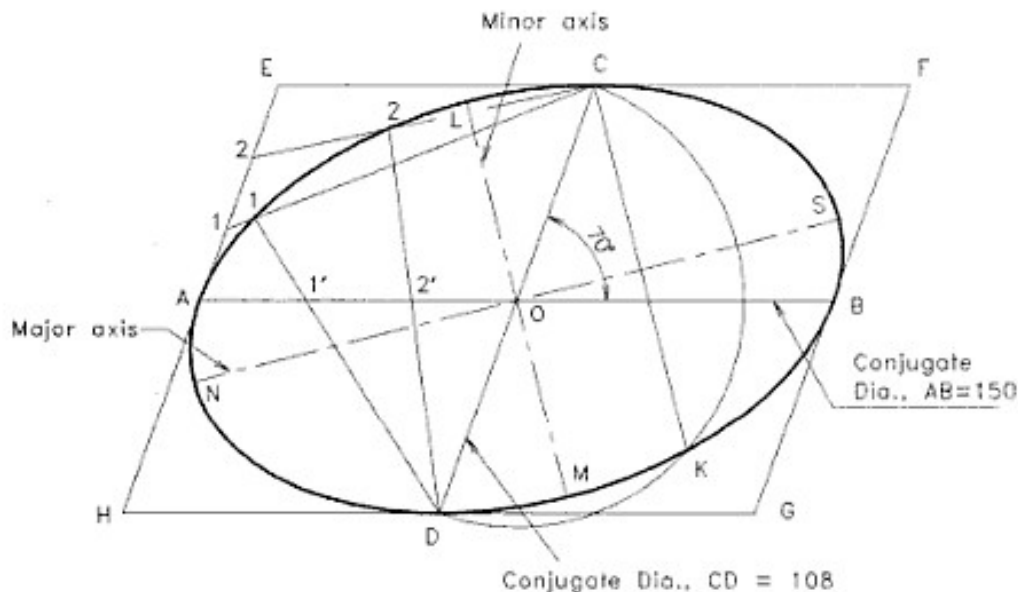


Fig. 6.8 Construction of an ellipse (parallelogram method—conjugate diameters given).

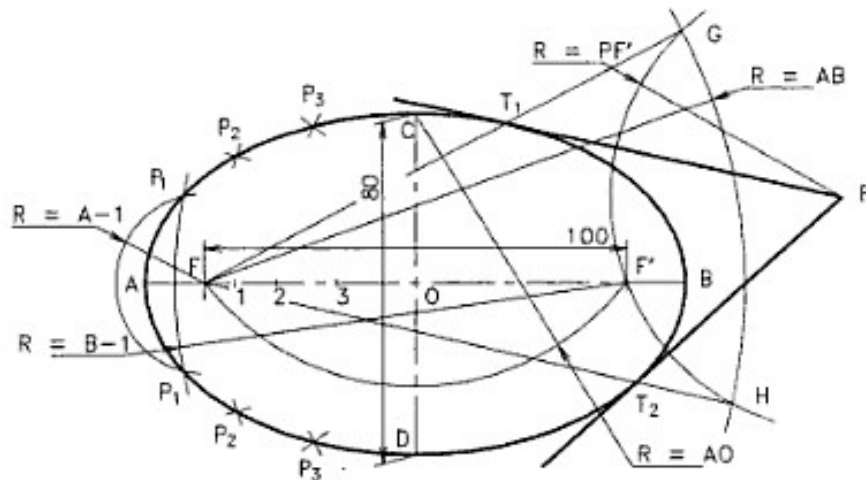


Fig. 6.9 Construction of an ellipse (intersecting-arc method).

8. Join FG and FH , intersecting the ellipse at T_1 and T_2 respectively. Join PT_1 and PT_2 . These are the required tangents.

Example 6.6

The distance between two coplanar fixed points is 100 mm. Trace the complete path of a point P moving in the same plane in such a way that the sum of the distances from the fixed points is always 152 mm. Draw the curve.

Refer to Fig. 6.10.

1. Draw a horizontal line and mark the fixed points F and F' in such a way $FF' = 100$ mm.
2. Draw perpendicular bisector CD . Mark the point C such that $FC = F'C = 152/2 = 76$ mm. Also mark the points A and B such that $AO = OB = FC = 76$ mm.
3. Mark a number of points on FO equal to the number of points required to be plotted in that portion of the ellipse. With F as centre and radius equal to AF draw an arc to cut the arc drawn with F' as centre and radius equal to $B1$. The point of intersection is marked as P_1 .
4. Similarly obtain other points of intersection P_2, P_3 , etc. It may be noted that $P_1F + P_1F' = P_2F + P_2F' = \dots = 152$ mm.
5. Draw a smooth curve passing through all these points to get the required ellipse.

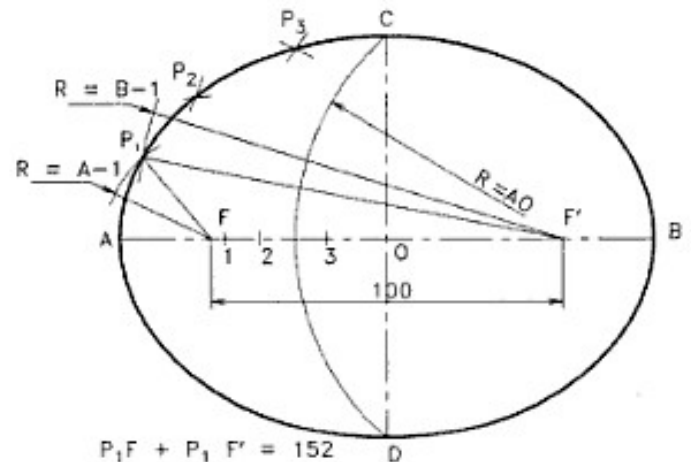


Fig. 6.10 Construction of an ellipse (given fixed points and the sum of distances or pin and thread method).

moving in a plane such that its distance from a fixed point called *focus* is always equal to its distance from a fixed line called *directrix*. A parabola has only one focus and one directrix. The terminology of parabola is shown in Fig. 6.11 and they are self explanatory.

Some of the important properties mentioned below will be useful in the construction of a parabola.

1. Area of the parabola is two-third the area of the circumscribing parallelogram.
2. Tangents PT and QT drawn from the extremities of any focal chord PQ intersect at T on the directrix at right angles.
3. Tangents drawn from the extremities of any chord intersect on the diameter which bisects that chord.

6.6 PARABOLA

A parabola may be defined as a curve traced out by a point

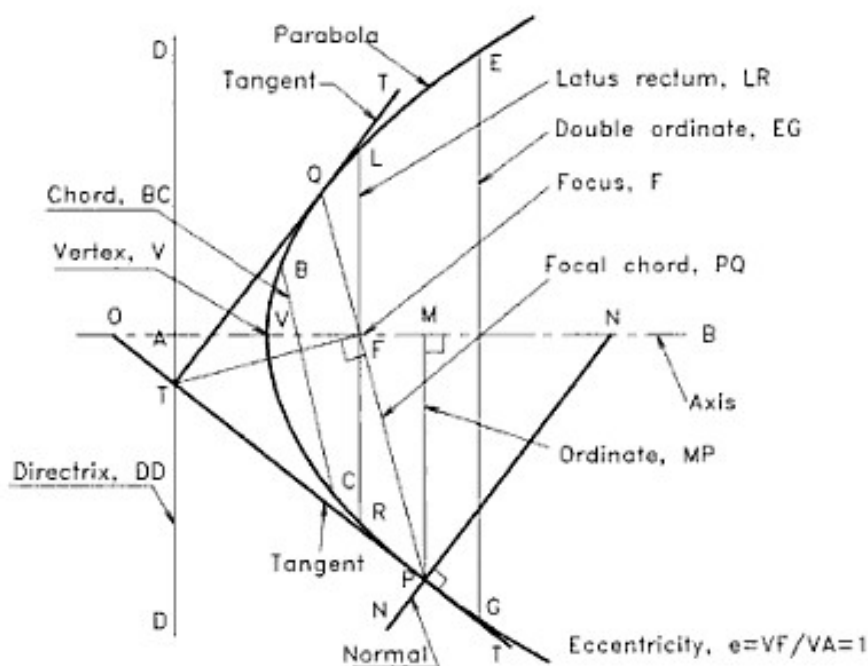


Fig. 6.11 Terminology of parabola.

4. Sub-tangent OM is bisected by the vertex of the parabola, V.
5. Sub-normal MN has a constant length and it is equal to twice the distance between the vertex and the focus.

Parabolic shapes are widely used in engineering practice. Some examples are head lamp reflector of automobiles, reflectors to concentrate solar power, bridge arches, cantilever bodies, shapes of machine tools, etc. The trajectory of missiles, the path of a jet of water, etc. are parabolic in shape.

6.7 METHODS OF CONSTRUCTION OF PARABOLA

A parabola can be constructed by various methods. A list of commonly used methods are given below.

1. Eccentricity method
2. Tangent method
3. Rectangular method
4. Parallelogram method
5. Offset method.

The following examples explain the five methods of construction of parabola and the drawing of tangent and normal to the curve.

Example 6.7

Construct a conic, when the distance between its focus and

directrix is equal to 50 mm and its eccentricity is one. Name the curve. Draw a tangent at any point on the curve.

Refer to Fig. 6.12.

1. As the eccentricity of the conic (e) = 1, the curve is a parabola. Draw the directrix DD and the axis AB

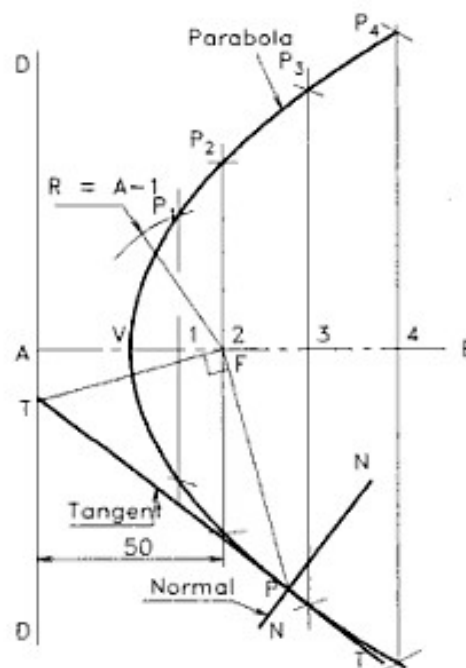


Fig. 6.12 Construction of an parabola (eccentricity method).

perpendicular to DD. Mark the focus F such that $AF = 50$ mm

- By definition $VF/VA = 1$ and hence mark the point V, the vertex at the midpoint of AF as shown in Fig. 6.12.
- Mark any number of points (say 4) on VB and draw perpendiculars through these points. With F as centre and AF as radius draw an arc to cut the perpendicular through point 1 at P_1 .
- Similarly, obtain points P_2, P_3, P_4 , etc. Draw a smooth curve passing through these points to obtain the required parabola.
- Tangent at any point P on the parabola can be drawn by joining FP and by drawing FT perpendicular to FP and extend. Draw NP perpendicular to TP. Now, TPT and NPN are the required tangent and normal at P.

Example 6.8

Draw a parabola by tangent method. Given its double ordinate = 120 mm and abscissa = 90 mm. Also locate the focus and directrix of the parabola.

Refer to Fig. 6.13.

- Draw the double ordinate $AB = 120$ mm and its abscissa $CV = 90$ mm.
- Extend the axis CV to E such that $CV = VE = 90$ mm. Join AE and BE and they are tangents to the parabola at A and B respectively.

- Divide AE and BE into the same number of equal parts (say 8). Join $1-1', 2-2'$, etc.
- Draw the curve tangential to all these line. This curve is the required parabola.
- Draw a line BJ parallel to EC. Locate the line BF such that $\angle JBE = \angle EBF$. Now, F is the focus of the parabola.
- With centre V and radius $= VF$ draw a semicircle to cut EC at G.
- Draw a line DD perpendicular to EC through the point G. Now DD represents the directrix.

Example 6.9

Draw a parabolic arc with a span of 1000 mm and a rise of 800 mm. Use rectangle method. Draw a tangent and normal at any point P on the curve.

Refer to Fig. 6.14.

- Draw an enclosing rectangle ABCD with base $AB = 1000$ mm and height $BC = 800$ mm using a suitable scale.
- Mark the axis VH of the parabola, where V is the vertex and midpoint of line CD.
- Divide DV and AD into the same number of equal parts (say 4).
- Draw a vertical line through the point $1'$ lying on the line DV. Join V with 1 lying on the line AD. These lines intersect at point P_1 as shown in Fig. 6.14. Similarly obtain other points P_2, P_3 , etc.
- Draw a smooth curve passing through these points to obtain the required parabola.

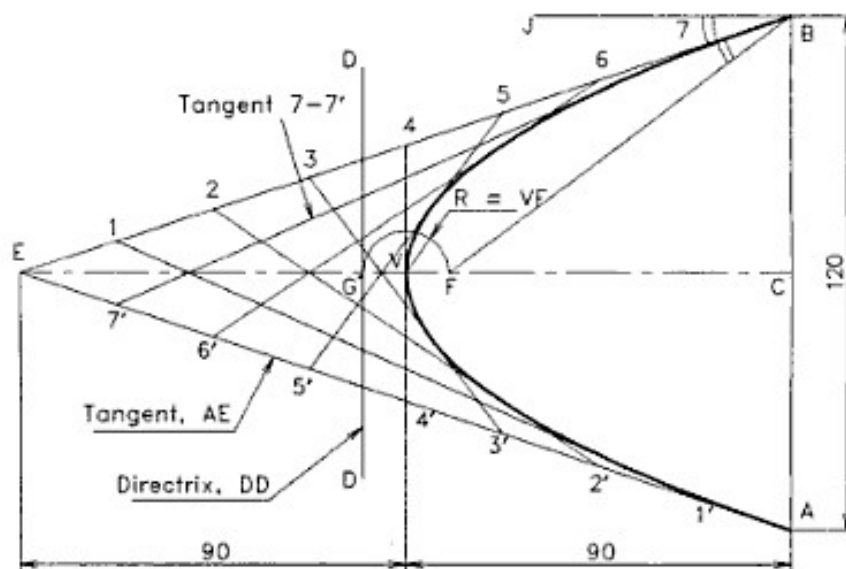


Fig. 6.13 Construction of a parabola (tangent method).

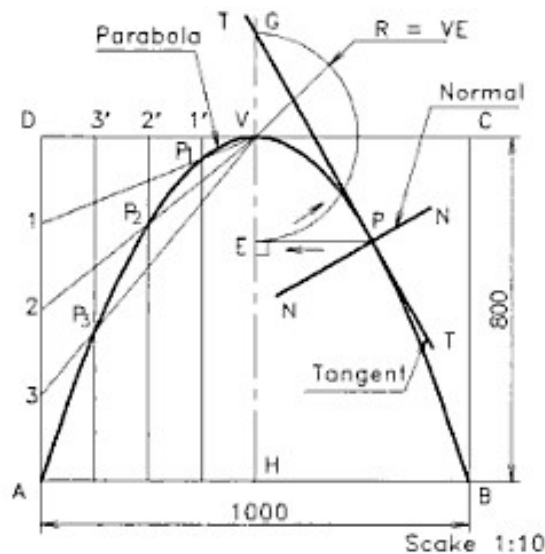


Fig. 6.14 Construction of a parabola (rectangle method).

- From point P draw the ordinate PE. With V as centre and VE as radius draw a semicircle to cut the axis produced at G.
- Join GP and extend it to T. Draw NP perpendicular to TP. Now, TPT and NPN are the required tangent and normal at P.

Example 6.10

Construct a parabola in a parallelogram of side 100 mm (chord) by 60 mm and with an included angle of 110° . Find the axis, focus and directrix.

Refer to Fig. 6.15.

- Draw the parallelogram KLMN as shown in Fig. 6.15.
- Join the midpoint O and P of the sides KN and ML.
- Divide the side KL and KO into equal number of divisions and join them as did in the Example 6.9 to get the points on the parabola.
- Repeat the same on the other half and draw a smooth curve through the points to complete the parabola.
- Draw MQ normal to OP and draw the perpendicular bisector AB to the line MQ. AB is the axis of the curve and V is the vertex.
- Mark a point T such that $VT = VE$ and join TM.
- Draw the line MF such that $\angle TMN = \angle TMF$. Point F gives the focus of the curve.
- Mark the point C on AB such that $CV = VF$. Draw DD perpendicular to AB through point C. DD is the direction of the parabola.

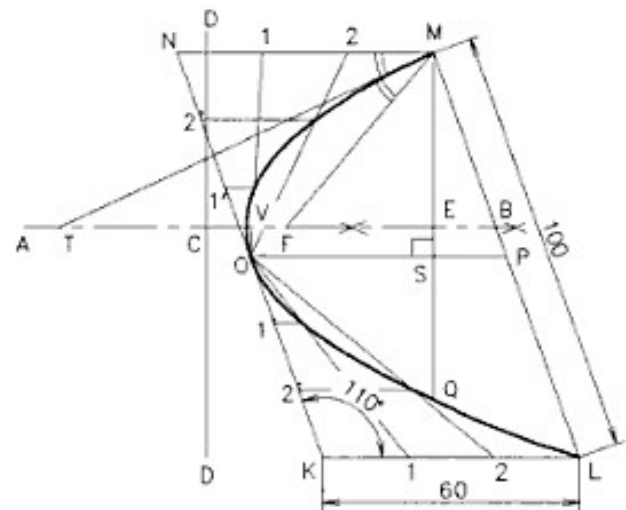


Fig. 6.15 Construction of a parabola (parallelogram method).

Example 6.11

Draw a parabola by offset method, given its base and axis as 160 mm and 96 mm respectively. Also draw tangents from any point Q outside the curve.

Refer to Fig. 6.16.

- Draw a rectangle KLMN such that $KL = 160$ mm and $LM = 96$ mm as shown in Fig. 6.16.
- Draw the perpendicular bisector AV of the line KL.
- Divide the side NV into any number of equal parts (say 4) and the side KN into the square of that ($4^2 = 16$).
- Draw horizontal lines from the divisions on the side KN at points $1' = 1$, $2' = 4$ and $3' = 9$. Also drop verticals from the divisions 1, 2, and 3 of the side NV to offset the point on the horizontals drawn such that $(1^2, 1)$ gives point P_1 , $(2^2, 2)$ gives P_2 and $(3^2, 3)$ gives P_3 on the required parabola.
- Repeat the same on the right half and complete the curve.
- Draw the line EG such that $BV = VG$. Then find the focus F by drawing the line EF keeping $\angle GE3 = \angle GEF$. Also locate the directrix DD by taking the length $FV = VC$.
- Let Q be the point outside the curve. To draw the tangents from Q, join Q to F and construct an arc at Q with radius QF to cut the directrix at points H and J.
- Draw perpendiculars to DD at H and J, to cut the parabola at T_1 and T_2 . Draw QT_1 and QT_2 to get the required tangents.

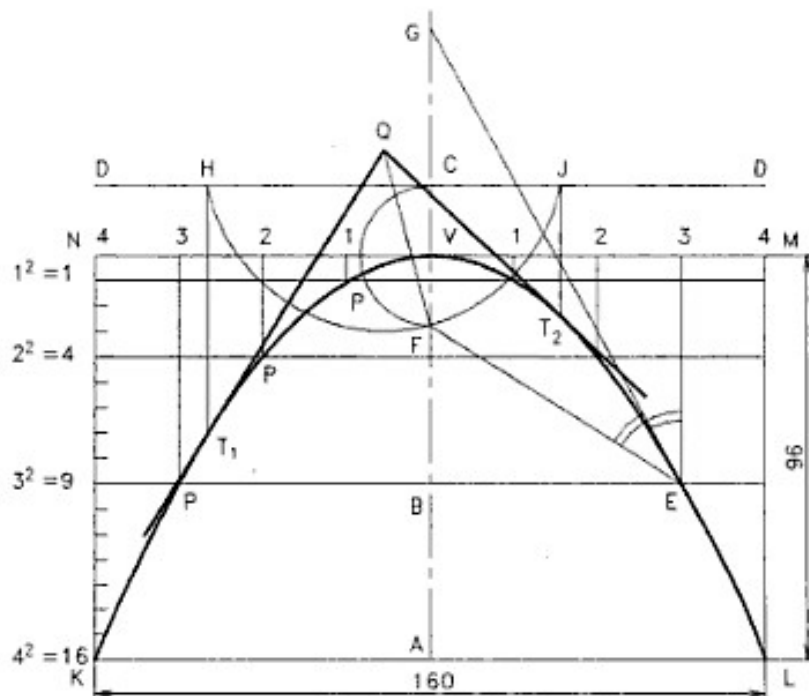
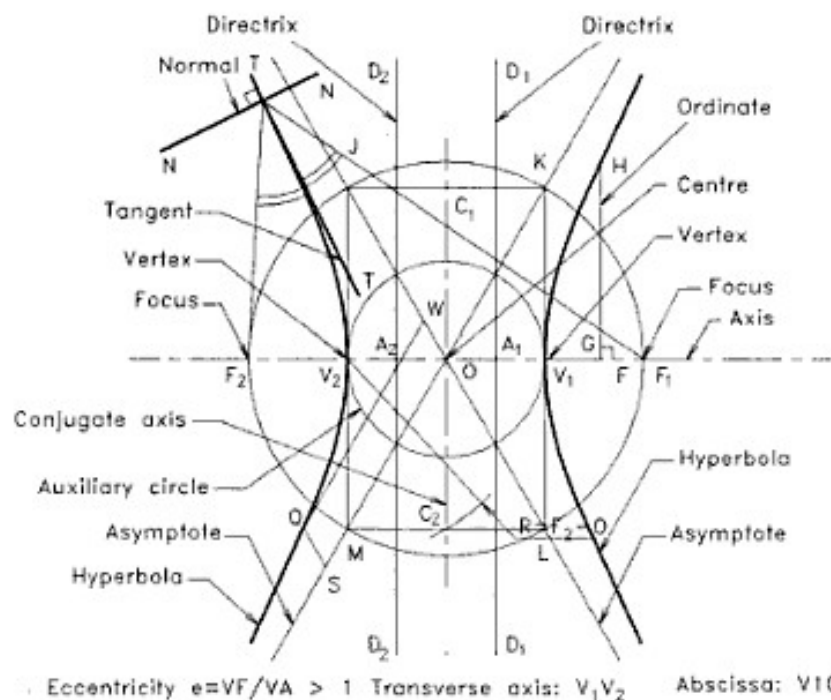


Fig. 6.16 Construction of a parabola (offset method).

6.8 HYPERBOLA

A hyperbola may be defined as a curve traced out by a point moving in a plane such that the difference between its distances from two fixed points called the *foci*, is a constant.

A hyperbola has two branches, two *foci* and two *directrices*. Each branch of the curve has the same eccentricity and hence hyperbola is also known as *central conic*. The terminology of hyperbola is shown in Fig. 6.17.



Eccentricity $e = VF/VA > 1$ Transverse axis: V_1V_2 Abscissa: V_1G

Fig. 6.17 Terminology of hyperbola.

The important terms associated with a hyperbola are defined below.

1. **Transverse axis:** It is the portion of the axis which lies between the two vertices V_1 and V_2 of the hyperbola.
2. **Centre:** It is the midpoint O of the transverse axis V_1V_2 .
3. **Conjugate axis:** It is the line C_1C_2 perpendicular to the transverse axis and passing through the midpoint O . The point C_2 can be located by cutting an arc with centre V_2 and radius equal to F_2O , on the perpendicular bisector of V_1V_2 .
4. **Auxiliary circle:** It is the circle drawn with midpoint O as centre and transverse axis V_1V_2 as diameter.
5. **Asymptote:** It is a tangent drawn at infinity to the curve. Actually, asymptote is a straight line which approaches the hyperbola without touching them, but it will touch the curves at infinity.

Asymptotes intersect the auxiliary circle on the directrix. Here, D_2D_2 and D_1D_1 are the directrices. The product of the distances of a point on a hyperbola from asymptote, one measured parallel to one asymptotes and the other measured parallel to the other asymptote is a constant; i.e. $QS \times QW = \text{constant}$.

It may be noted that tangent at any point P on the curve bisects the angle F_2PF_1 i.e. angle F_2PT is equal to angle F_1PT . Here, MN is the normal.

A rectangular or equilateral hyperbola may be defined as the locus of a point moving in such a way that the product of its distances from two fixed lines at right angle to each other is constant. These fixed lines are the asymptotes to the rectangular hyperbola.

6.9 METHODS OF CONSTRUCTION OF HYPERBOLA

Similar to parabola, hyperbola also can be constructed by various methods. A list of commonly used methods are given below.

1. Eccentricity (focus and directrix) method.
2. Foci and the difference of the distances method.
3. Ordinate, abscissa and transverse axis method.
4. Asymptotes and a point on the curve method.

The following examples explain the four methods of construction of hyperbola and the drawing of tangent and normal to the curve.

Example 6.12

Construct a conic, when the distance of any point P between the focus and the directrix is constant and is equal to $3/2$.

Name the curve. Draw a tangent and normal at any point on the curve. The focus is located 50 mm away from the directrix.

Refer to Fig. 6.18.

1. As the eccentricity is greater than 1, i.e. $e = 3/2 > 1$, the curve is a hyperbola. Draw one directrix DD and the axis AB perpendicular to DD and mark the focus F such that $FA = 50$ mm. As the eccentricity is $3/2$, divide FA into $3 + 2 = 5$ equal parts. By definition $VF/VA = 3/2$ and hence locate the vertex V .

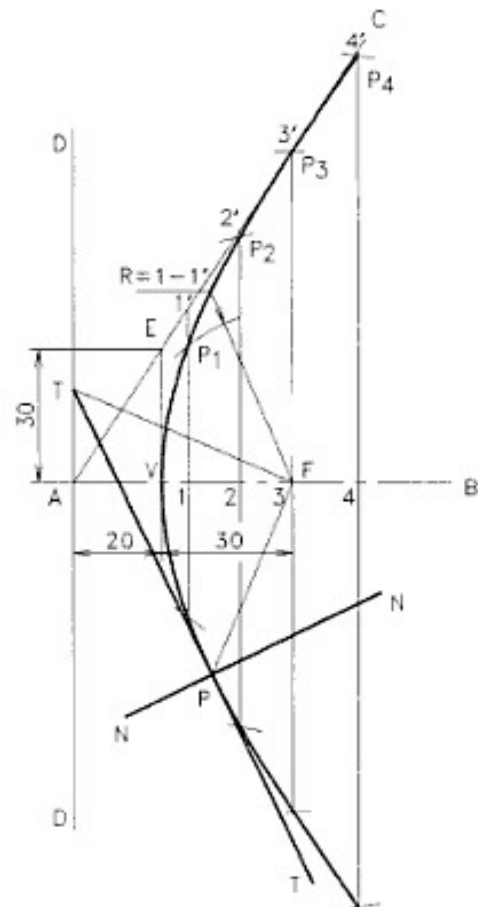


Fig. 6.18 Construction of a hyperbola (given focus and directrix).

3. Draw perpendicular to the axis such that $VE = VF$. Join AE and extend it as shown in Fig. 6.18. This is the eccentricity scale, which gives the distances in the required ratio. In triangle AVE , $EF/VA = VF/VA = 3/2$.
4. Mark any point 1 on the axis and proceed further as explained in Example 6.7 to get the points P_1, P_2, P_3 , etc. which is the required hyperbola.
5. Tangent and normal at any point P on the hyperbola can be drawn as shown in the Fig. 6.18.

Example 6.13

Two points F_1 and F_2 are located on a plane sheet 100 mm apart. A point P on the curve moves such that the difference of its distances from F_1 and F_2 always remains 50 mm. Find the locus of the point and name the curve. Mark asymptotes and directrices.

Refer to Fig. 6.19.

1. A curve traced out by a point moving in the same plane in such a way that the difference of the distance from two fixed points is constant, is called a *hyperbola*. Draw a horizontal line and mark the fixed points F_2 and F_1 in such a way that $F_2F_1 = 100$ mm.
2. Draw a perpendicular bisector C_1OC_2 to F_2F_1 as shown in Fig. 6.19. Mark the points V_2 and V_1 on the horizontal line such that $V_2V_1 = 50$ mm and $V_2O = V_1O$.

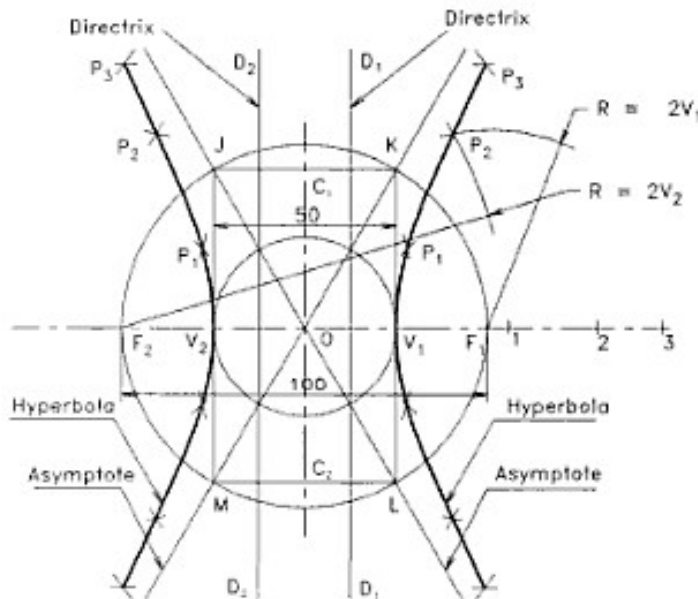


Fig. 6.19 Construction of a hyperbola (given fixed points and the difference of the distances).

3. With centre O and radius equal to F_2O draw a circle.
4. Draw tangents at V_2 and V_1 to intersect the above circle at J, M, K and L as shown.
5. Draw a line joining JOL and produce it and this line is one asymptote. The other asymptote is the line passing through KOM .
6. Mark any number of points 2, 3, 4, etc. on the axis of the hyperbola. With F_2 as centre and radius equal to $2V_2$ draw an arc to cut the arc drawn with F_1 as centre and radius equal to $2V_1$. The point of intersection is

marked as P_2 . Similarly, obtain other points of intersection P_3, P_4 , etc.

7. It may be noted that the differences $P_2F_2 - P_2F_1 = P_3F_2 - P_3F_1 = \dots = 50$ mm. Draw a smooth curve passing through the points V, P_1, P_2, P_3 , etc. which is the required hyperbola. Also draw another hyperbola on the other side of the axis, symmetrical about C_1 and C_2 as shown in the Fig. 6.19.

Example 6.14

Draw a hyperbola, given the transverse axis = 80 mm, single ordinate = 60 mm and abscissa = 48 mm.

Refer to Fig. 6.20.

1. Draw a horizontal line V_2V_1 of length 80 mm to represent the transverse axis and extend it to E such that V_1E is equal to 48 mm. V_1E represents the abscissa.
2. Draw a double ordinate, BEC of length two times that of the ordinate (i.e. 2×60 mm) perpendicular to the axis. Complete the rectangle $ABCD$. Divide the sides DC and CE into the same number of equal parts (say 4) as shown in Fig. 6.20.

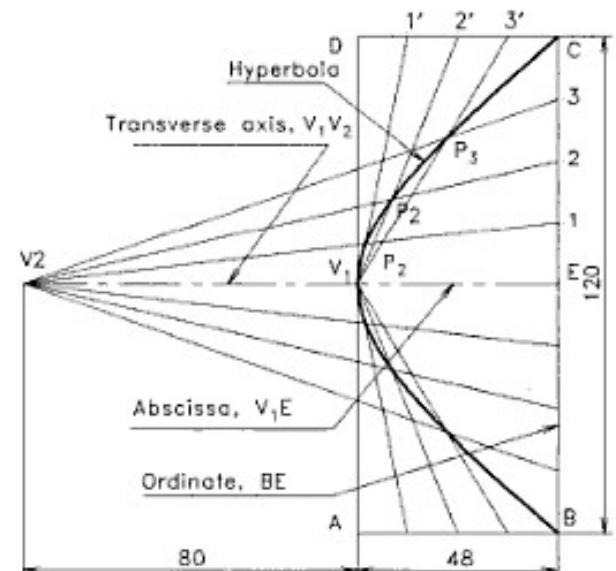


Fig. 6.20 Construction of a hyperbola (given ordinate, abscissa and transverse axis).

3. Join V_1 with $1', 2'$ and $3'$. Also join V_2 with 1, 2 and 3 to get the intersecting points P_1, P_2 and P_3 respectively.
4. Similarly, get the intersecting points on the other side of the axis. Draw a smooth curve passing through all these points which gives the required hyperbola.

Example 6.15

The asymptotes of a hyperbola make 80° with each other. A point P on the curve is at a distance of 30 mm from the horizontal asymptote and 45 mm from the inclined asymptote when measured horizontally. Plot the hyperbola. Draw tangent and normal at any point M on the curve.

Refer to Fig. 6.21.

1. Draw the asymptotes OX and OY making an angle of 80° with each other as shown in Fig. 6.21. Mark the given point P. Through point P draw line AB and CD parallel to OX and OY respectively. Mark any point 1 on the line PC and join O1 which intersects the line AP at 1'.

2. Through points 1 and 1' draw lines $1P_1$ and $1'P_1$ parallel to OX and OY respectively to intersect at P_1 . Similarly, obtain points P_2, P_3, P_4 , etc. after marking points 2, 3, 4, etc. on both lines PC and PB.
3. Draw a smooth curve passing through all these points to give the required hyperbola.
4. To draw a tangent at any point M on the curve mark point M and complete the parallelogram OLMK.
5. Mark the point T_1 on OY such that $OT_1 = 2 \times OL$ and mark T_2 on OX such that $OT_2 = 2 \times OK$.
6. Join T_1 and T_2 through the points M. Now, NMN is the required normal at M.

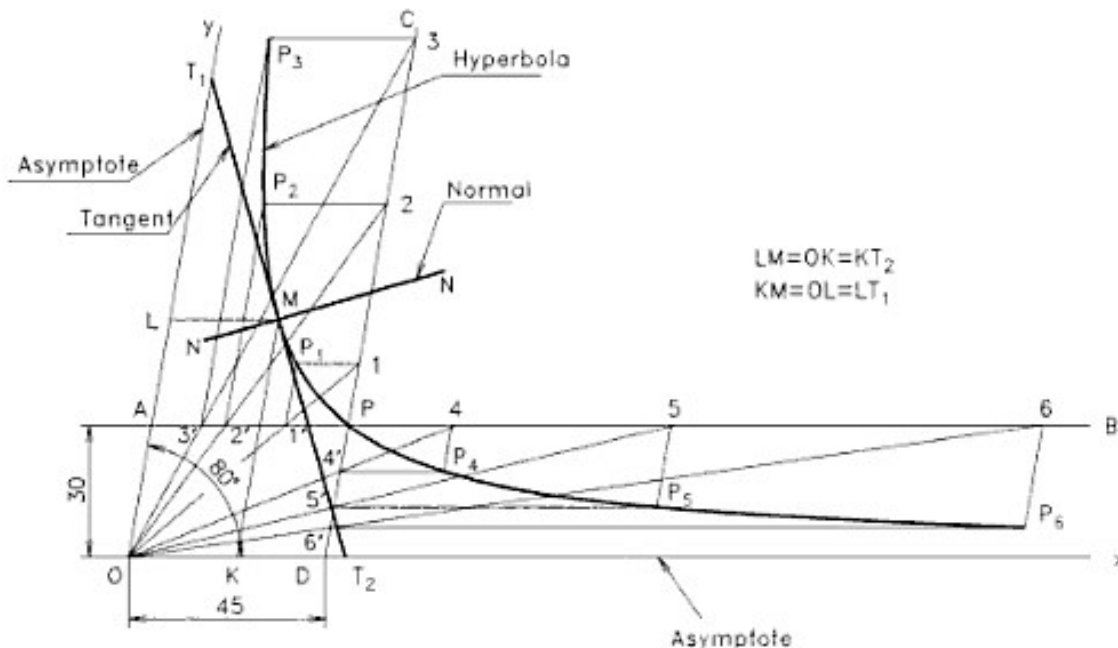


Fig. 6.21 Construction of a hyperbola (given asymptotes and a point on the curve).

EXERCISES**Ellipse**

1. The focus of an ellipse is 50 mm away from its directrix. If the eccentricity is $\frac{2}{3}$, draw the curve and measure its major and minor axes. A point P is located 45 mm from the second vertex. Draw tangent and normal to the curve at that point.
2. The major and minor axes of an ellipse are 100 mm and 70 mm respectively. Construct the curve using the method of auxiliary circles. Draw a tangent and normal at a point P on the curve located at 35 mm from the top end of the minor axis, towards left.
3. Draw the largest possible ellipse in a rectangle of 120×80 mm and mark its focus and axes using the oblong method. A point P on the lower half of the curve is at a distance of 20 mm from the left end of the major axis. Mark the point and draw tangent to the curve at that point.
4. The conjugate diameters of the ellipse are 130 mm and 90 mm with an included angle 65° . Draw the ellipse by parallelogram method and determine its axes.
5. The foci of an ellipse are 100 mm apart and the minor

axis is 70 mm long. Draw an ellipse by intersecting arc method. A point P is located outside the curve at a distance of 80 mm from the bottom end of the minor axis and 20 mm from the right end of the major axis. Draw tangents to the curve from the point P.

6. The distance between two coplanar fixed points is 110 mm. Trace the complete path of a point P moving in the same plane in such a way that the sum of the distances from the fixed point is always 150 mm. Draw the curve and a tangent at any point on the curve located at the right top quarter.

Parabola

7. Construct a conic when the distance between its focus and directrix is equal to 40 mm and its eccentricity is one. Draw a tangent at a point on the upper half of the curve located 60 mm from the focus.
8. The double ordinate of a parabola is 110 mm and abscissa is 80 mm. Draw the parabola and locate its focus and directrix.
9. The span and rise of a parabola are 1200 mm and 880 mm respectively. Using rectangle method, construct the parabola and draw tangent and normal to the curve at a point on it, located 30 mm from the vertex on the right side.
10. Construct a parabola in a parallelogram of side 100 mm (chord) $\times$ 60 mm and with an inclined angle of 120° . Find the axis, focus and directrix of the curve.

11. By offset method, construct a parabola having base equal to 120 mm and axis 112 mm. Draw the tangents to the curve from a point P outside the curve such that the point P is located 20 mm towards the left of the axis and 5 mm below the directrix.

Hyperbola

12. Construct a conic when the eccentricity is $5/4$ and the distance between directrix and the focus is 54 mm. Draw tangent and normal to the curve at a point on the lower half of the curve measuring 45 mm from the focus.
13. The foci of the two branches of a hyperbola are located at a distance of 80 mm. A point P on the curve moves such that the difference of its distances from the foci always remain 40 mm. Find the focus of the two branches of the curve, mark the asymptotes and measure the angle between them.
14. Draw a hyperbola when the transverse axis is equal to 90 mm, double ordinate = 144 mm and abscissa is equal to 48 mm.
15. The asymptotes of a hyperbola are at an angle of 75° . A point P on the curve is 25 mm away from the horizontal asymptote and 40 mm away from the inclined asymptote, when measured horizontally. Construct the curve and draw tangent and normal at a point M on the curve, 50 mm above the horizontal asymptote.